

Screen Time, Two Deletions

AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7) · TEACHER KEY

CED TETHER

- VAR-1.A / UNC-1.A: identify the question to be answered and the individuals and variables in a context.
- VAR-1.B: classify variables as categorical or quantitative; a label or identifier is not a measurement.
- UNC-1.H / 1.I: compare groups of unequal size with relative frequencies, not raw counts.
- UNC-1.J: describe a distribution with shape, center, variability, and unusual features.
- UNC-1.K: the median is resistant to unusual values; the mean is not, so report the center that fits the data.

Essential question: What does a number mean, and what changes when you delete the values you do not like?

DOK-3 skill: 4.B · **FRQ pattern:** judge-a-trimmed-report-using-context-resistance-and-relative-frequency

DOK rationale: Part (d) requires a defended decision: whether a report that deleted two values is responsible, which statistic to report, and what the changed percent means for the principal’s actual question. It coordinates context, individuals-versus-variables, relative frequency, and resistance rather than recalling any one of them.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (d) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) and (c) are the ladder up; use them to see where a thin (d) came from.

Questions to ask

- Who exactly was measured, and what is one row of the data?
- Which of your variables could you add up, and which could you only count?
- If the two groups were the same size, would your comparison change?
- Which number moved a lot and which barely moved when the two values were deleted? Why?

[WARN] LOOK FOR

- Writing a number with no who/what/units (the 30% left floating).
- Calling the student ID or the device name a quantitative variable.
- Comparing 9 freshmen to 3 seniors by counts and declaring freshmen three times worse.
- Saying the mean is resistant, or that deleting the two values was fine because the median barely moved.
- Describing the distribution without mentioning the two unusual values at all.

SCORING PART (d) — E / P / I

Expected elements

- **e1** (required) — Gives the 26% its context: who (students in the sample), what (daily screen time over 240 minutes), and the question it answers.
- **e2** (required) — Chooses a center and justifies it with resistance, consistent with part (c).
- **e3** (required) — Interprets the big mean change versus the small median change as the effect of the two unusual values.
- **e4** (required) — States that the deletion must be disclosed and justified (logging errors) and that unusual values still belong in the description.

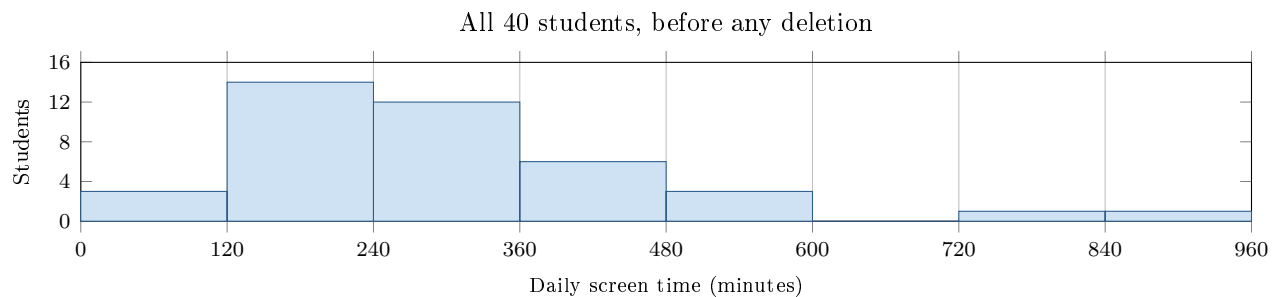
E	All four elements, in the context's words, with a clear decision.
P	A decision with two or three elements, or all four with the context or the disclosure vague.
I	A bare number or a decision without resistance, context, or the disclosure.

[STAR] THE PROBLEM — annotated

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*



FIRST TAKE — one sentence before you work the parts

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

Key: Ungraded commitment; accept any position that names something the first sentence leaves out.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.

Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:		
Variable	One value looks like	Categorical / Quantitative
Grade level		
Device used most		
Daily screen time		

Key: Individuals: the 40 students in the sample. Grade level: one value is Freshman or Senior, categorical. Device used most: one value is Phone, Laptop, or Console, categorical. Daily screen time: one value is a number of minutes such as 176, quantitative (it can be averaged).

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: *Of the _____ freshmen, _____ are over 240, which is _____%. Of the _____ seniors, _____ are over 240, which is _____%. So the fair comparison is _____ because the groups _____.* Then explain in one sentence why the count 9 versus 3 can mislead here.

Key: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. The fair comparison is 36% versus 20% because the groups have different sizes; 9 versus 3 exaggerates the gap (three times as many students, not three times as bad), and freshmen are still higher but by less than the counts suggest.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

Key: The median is resistant; the mean is not. The two unusual values at 720 and 900 minutes sit far to the right of the rest of the distribution, so they pull the mean upward. Removing them drops the mean by 27 minutes, while the median, which depends only on the middle of the ordered data, moves by 2.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

Key: Report the median (about 174–176 minutes) because it is resistant and describes the typical student either way. Removing the two values changed the mean a lot but barely changed the median, which shows the mean was being pulled by two unusual values, not by typical students. The 26% means that 10 of the 38 remaining students in the sample reported more than 240 minutes per day. Deleting the two entries is defensible because they were logging errors, not real days, but the report must say that two values were removed and why, and it should still note that unusual values existed; only then does the number connect to the principal's question about whether screen time is a concern.

Word bank: **median** · it resists unusual values · mode · **mean** · it follows unusual values
· omit the deletion details · **disclose and justify the deletions**

bold = needed; the rest are distractors